ABM apply to Santa Lucia, Colombia

August 2026

1 Demographic and economic calibration

Parameter	Value used	Source
Simulated population (N)	972 people	
Age structure	27.6% (0–14), 64.4% (15–64), 8.0% (65+)	DANE population projection, Santa Lucía, 2026
Gender split	49.6% male / 50.4% female	Same source
Ethnic self-identification	4.4% Indigenous, 6.8% Afro-Colombian/Black/Mulatto, 88.8% none declared	DANE 2018 Census
Literacy (risk-perception proxy)	93.4% "full comprehension"	Departmental illiteracy rate, DANE 2005 Census
Income classes (monthly per capita, COP)	Poor 0–435,375; Vulnerable 435,375–853,608; Middle 853,608–4,596,352; Upper 4,596,352+	DANE, 2024 Departmental Monetary Poverty
Sectoral economic constants	Scaled by Colombia/Canada GDP per capita × COP/USD exchange rate	World Bank 2024; Banco de la República

Table 1: Values used to adapt the model

2 Experiment 1: community organization

This experiment asks if a community invests in organizing itself before a flood (something like the *Juntas de Acción Comunal* model of local organizing common in Colombia), how much does that change flood outcomes? This is controlled in the model by a parameter, community_organization_level, which ranges continuously from 0 (no organization) to 1 (full organization). Five discrete levels were simulated: 0, 0.25, 0.5, 0.75, and 1.0, with $n = 10$ independent runs at each level. This parameter implements two functionalities:

Direct flood-depth attenuation When a person's position is inside a flooded area, the model draws a random flood depth, then reduces it according to the organization level:

$$\text{raw_depth} \sim \mathcal{U}(10, 55) \text{ cm}, \qquad \text{depth}_{\text{effective}} = \text{raw_depth} \times \big(1 - c_{\text{depth}} \times \text{org_level}\big) \tag{1}$$

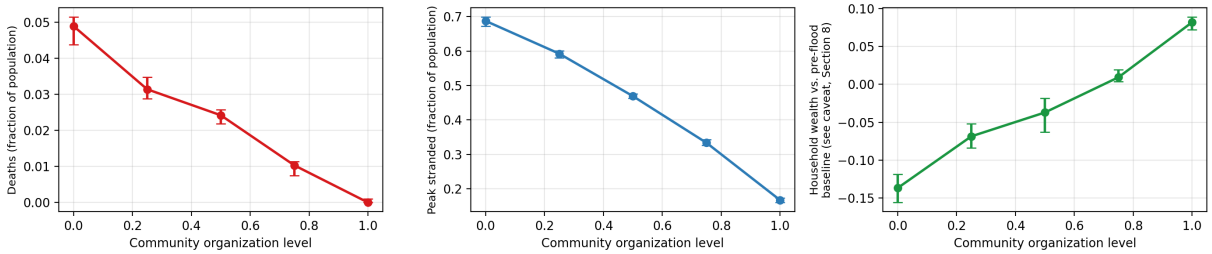
where c_{depth} is a coefficient defaulting to 0.5 (its a model assumption) . At full organization (org_level = 1.0) with the default coefficient, this halves the flood depth every agent experiences – representing organized, collective flood-fighting (reinforcing embankments, blocking culverts, standing watch), of the kind documented for the neighboring town of Suán during the real 2010 event, in contrast to Santa Lucía's own improvised defenses, which failed [1].

Social-capital boost. Independently of flood depth, each agent's three social-capital components (bonding, bridging, and linking ties, used by the model's Social Cognitive Theory decision rule) are boosted:

$$\text{community_boost} = \text{org_level} \times c_{\text{sct}}$$
$$\text{bonding_count} = \min\big(1,\ \text{random}(0,1) + \text{community_boost}\big)$$

with c_{sct} defaulting to 0.3. This is meant to represent organized communities having denser social networks and more mutual trust, which in principle should translate into more people acting on flood warnings and helping each other.

2.1 Results



Organization level	Deaths	Peak stranded	Household wealth
0.00	4.9%	69%	−13.7%
0.25	3.1%	60%	−6.6%
0.50	2.4%	47%	−3.6%
0.75	1.0%	33%	+1.3%
1.00	0.0%	20%	+8.2%

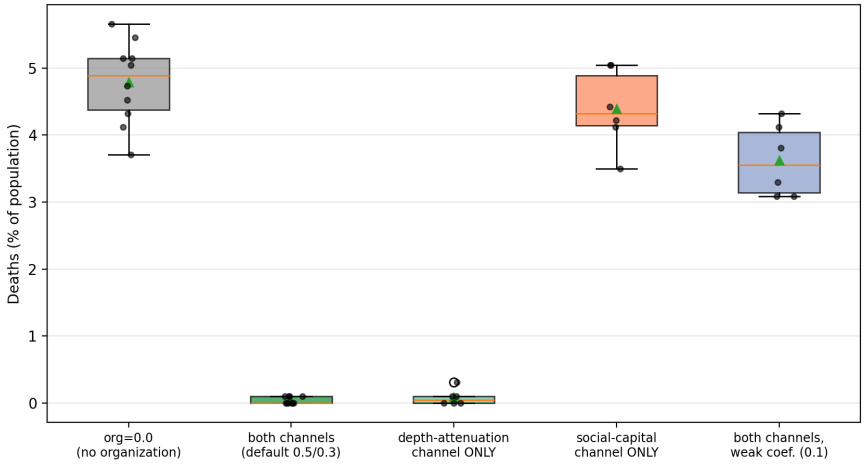
Table 2: Median outcomes across the 5 organization levels, $n = 10$ runs.

2.2 Testing each channel independently

Holding organization level fixed at 1.0 (full organization), five configurations were run:

- **org=0.0 (no organization)**.
- **both channels (default coefficients 0.5 / 0.3)**.
- **depth-attenuation channel only** – social-capital forced off.
- **social-capital channel only** – depth attenuation forced off.
- **both channels, weak depth coefficient (0.1)** – both channels active.

2.3 Results



This figure compares whether the links formed between people and the information networks that emerge from these interactions are more relevant, or whether depth attenuation is the more significant factor.

3 Shelter capacity, distance, and rescue fleet

3.1 What is being tested

The original model's rescue logic only checked whether the shelter had a free space; once a spot opened up, a stranded person was placed in the shelter instantly, regardless of how far away it actually was. This experiment replaces this with a distance-aware and fleet-aware rescue model, and tests two policy levers together: how big the shelter is (its capacity, as a percentage of the population it can hold) and how far it is from the flooded area, using four real candidate shelter locations with distances computed by real-world routing rather than straight-line distance:

Site	Real routed distance
Nearby crossing	2.0 km (pure water crossing)
Hypothetical location	38.4 km
Intermediate site	58.0 km
Barranquilla	99.9 km

These distances were computed with OSRM (Open Source Routing Machine), following real road/water routes rather than straight-line distance, which would understate actual travel time.

Step 1: How long one round trip takes. If the route is a pure water crossing (true only for the 2.0 km "nearby crossing" site), travel time uses a boat speed of 5 km/h (deliberately slow, representing flood water full of current and debris). For longer routes that combine a water crossing near the flood zone with dry road once clear of it, the first 1.5 km is costed at boat speed and the remainder at a dry-road speed of 60 km/h – this 60 km/h figure is not an assumption but the empirical average of three real OSRM-routed trip durations for the shelter candidates, back-calculated from real distance and time:

$$\text{one_way_hours} = \begin{cases} \dfrac{\text{distance}_{\text{km}}}{5} & \text{pure water crossing} \\ \dfrac{\min(\text{distance}_{\text{km}},\ 1.5)}{5} + \dfrac{\max(0,\ \text{distance}_{\text{km}} - 1.5)}{60} & \text{mixed route} \end{cases} \tag{2}$$

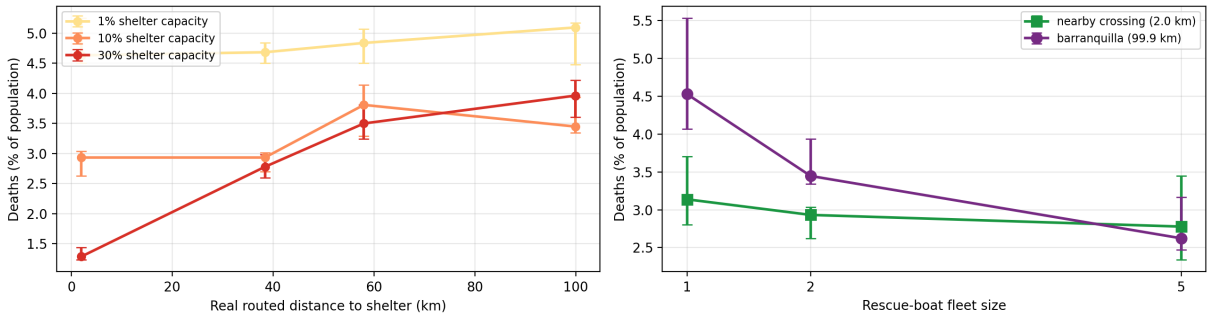
$$\text{round_trip_hours} = \max\big(2 \times \text{one_way_hours},\ 0.2\big) \tag{3}$$

(the 0.2h / 12-minute floor represents loading and unloading time, applied even at zero distance).
Step 2 – how many people the fleet can move per hour. With rescue_boats_available boats (default: 2) each carrying boat_capacity_per_trip people per trip (calibration default: 6):

$$\text{throughput}_{\text{people/hour}} = \frac{\text{boats} \times \text{capacity per trip}}{\text{round_trip_hours}} \tag{4}$$

Step 3 – enforcing the bottleneck every simulated hour. A stranded agent is only moved into the shelter if *both* conditions hold: the shelter still has a free space *and* the fleet has not already used up its hourly throughput budget. If the hourly throughput budget is exhausted, the agent stays stranded for at least one more hour, even if the shelter itself still has room. This is what makes distance causally matter for the first time in the model: farther shelters mean longer round trips, which mean a lower hourly throughput, which means more hours spent stranded, for a fixed fleet size.

3.2 Results



This graph has two parts. The one on the left shows how important the distance to the shelter is, depending on how large the shelter is. When the shelter is very small (yellow line, 1% capacity), the distance barely matters: it fills up so quickly that all locations end up with similar results. But when the shelter has more space (red line, 30% capacity), the effect becomes noticeable: the farther away the shelter is, the more people die.
The one on the right shows what happens when the number of available rescue boats is changed. At the closest site (green line), having more or fewer boats makes almost no difference because the trip is so short that one boat is already enough. But at the farthest site—Barranquilla (purple line)—adding more boats makes a big difference: with only one boat, mortality is high, and it drops noticeably when there are two or five boats.

References

[1] El Heraldo. *Santa Lucía y Suan: los contrastes de la catástrofe*. 2020. URL: `https://www.elheraldo.co/atlantico/santa-lucia-y-suan-los-contrastes-de-la-catastrofe-776688` (visited on 08/21/2026).